Journal Paper Summary (25 points total)

Paper 1 Title	Letter-value plots: Boxplots for large data
Author Name(s)	Heike Hofmann, et al.
Paper 2 Title	Opening the Box of a Boxplot
Author Name(s)	Yoav Benjamini
Student Name	Rey Stone
Student ID	111044637

1. What is a boxplot? Find and discuss the limitations/ challenges listed with respect to the use of boxplots across both papers. Discuss how other variations of the boxplot help overcome them. [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the paper) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts) • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments

Write your answer below:

According to Hofmann et al, a boxplot is a type of graph that can give a "compact graphical summary of the distribution of a variable". The shortcomings of boxplots include 1) the number of "outliers" that grows linearly with sample size and 2) tail behavior estimates are not displayed. Having too many outliers in the data may be genuine but extreme, or otherwise true outliers from a different distribution. Hofmann argues that the boxplots tend to display "too many outliers" when looking at Gaussian data. Hofmann cites Benjamini's paper, noting that vase and box-percentile plots are good ways to provide more detailed information for larger sample sizes, while acknowledging that they depend on specifically a kernel density estimate and additional smoothing. Hofmann's own paper dives into letter-value plots which proves more accurate for larger sample sizes and fixes the tail

behavior of box plots by using letter values to define outliers according to a defined “ordering of statistics that have specific depths”.

According to Benjamini a boxplot is a power tool for displaying data, summarizing symmetry, “longtailedness”, and other features of a “single batch of data”. Benjamini mentions that some of the shortcomings of the boxplot are a “compromise between a detailed description of batch data and a condensed display of summaries”. Benjamini’s alternatives are stem-and-leaf displays, histograms, density traces, but Benjamini also proposes the vaseplot which is where the width of a boxplot indicates the estimated density (making use of the width of boxplots since he posits earlier “the only question I sometimes get is : How wide should the box be?”).

2. List the reasons why boxplots have become popular or useful as mentioned per paper i.e. mention the benefits, usefulness or reasons why the boxplot is popular, citing the relevant paper. [5 points]

Focus of this question
• This question encourages you to connect your learning to "real world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

From the second paper literally in the first section:

1. Five data summaries are easily presented right in front of your face.
2. The boxplot displays detailed information about the observations at the tails
3. Distributions of many batches can be easily compared just by side-by-side
4. Boxplots have detailed information at the table
5. They are easy to compute, and they can be easily explained

From the first paper:

- They are useful displays for conveying rough information about the central 50% and the extent of data.
- They give a compact graphical summary of the distribution of a variable.
- Conventional boxplots rely on the fourths to "minimize the difficulties of masking" and because the rule is then "unlikely to be adversely affected by extreme observations".

What is a letter-value plot according to the papers? Do you think it is an improvement over the traditional boxplot? Why or why not? Do you agree with Tufte (1983) recommendation about the boxplot? Why or why not? [15 points]

Focus of this question
• This question encourages you to connect your learning to "real world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to reflect on what you are learning • This question encourages you to contribute your ideas and thoughts • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments • This question challenges you to develop and present your own knowledge, comprehension, and conceptual understanding

Write your answer below:

According to the paper, "letter-value plots are based on the letter values, with one box for each pair of lower and upper letter values". They add onto box plots by incrementally narrowing the boxiness of the plot showing the data spread and shading the inner boxes heavily to indicate density. This is a huge improvement off traditional boxplots because it keeps the simplicity of the summary of data one can glean from a box plot while adding other statistics with little friction into the mix. The outlier problem is "solved" with the narrowing of the width of the box plots, with only the extreme tails being dots.

Tufte has several recommendations for the box plot. In Benjamini's paper, he recommends "stripping down the boxplot to its essentials. He finds no use for the box, as only its upper and lower positions are needed." I disagree with this. I think that knowing the interquartile range of the data is extremely useful and seeing where the median and average fall can be a source of information.